

Microwave Rectifiers with Feedback and Adaptive Strategy for Large Dynamic Input Power Levels

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Abstract—Rectifiers are crucial for wireless power transmission and energy harvesting. This paper addresses the challenges and solutions for enhancing conversion efficiency across various input power levels. We present the advancements in rectifiers for wireless power transmission and discuss future applications. Critical requirements for rectifiers are outlined alongside technologies for achieving high efficiency at varying power levels. Strategies to reduce harmonic and diode losses at various power levels, as well as adaptive rectifying techniques, are also proposed. We suggest rectifier designs that maintain high efficiency across a wide input power dynamic range.

Keywords—rectifier; rectenna; wireless power transmission; high efficiency

I. INTRODUCTION

Since Nikola Tesla first introduced electromagnetic wireless power transmission (WPT) over a century ago, the technology has significantly progressed. In 2007, a team from MIT successfully powered a 60-watt light bulb from two meters away using magnetic resonance WPT, a breakthrough that was published in Science [1]. This demonstration reignited interest in WPT, which now serves crucial functions in various areas, including space solar power systems (SSPS) [2], energy transfer between satellites [3], stratospheric solar power stations [4], and the wireless power supply for unmanned aerial vehicles (UAVs) during flight [5]. There are a lot of studies on microwave rectifiers that present various topologies with high efficiency. [6][7][8][9] The potential of WPT is vast, and there is inspiring hope for its future applications in areas we have yet to explore.

A microwave wireless power transmission system consists of a microwave source, a transmitting antenna, a receiving antenna, and a rectifying circuit. Enhancing the efficiency of each component is essential for improving the overall efficiency of the wireless power transmission system. The rectifying circuit is a nonlinear circuit and serves as the key device for converting microwaves into direct current. In certain scenarios, such as wireless power transmission to moving targets, the power received at the receiving end can vary significantly with distance. Therefore, the microwave rectifying circuit at the receiving end must possess a certain power dynamic range to achieve

high rectification efficiency across various power levels.

Rectifying devices are nonlinear components, and their impedance changes with varying microwave power, leading to a decrease in rectification efficiency. Designing impedance matching circuits can expand the range of microwave power variations. By employing dynamic compensation techniques, variable-capacitance diodes can help adjust the impedance matching range. Additionally, utilizing adaptive technology allows for the automatic selection of the rectifying circuit based on the level of input microwave power, enabling the maintenance of higher rectification efficiency across a broader power dynamic range.

II. FEEDBACK TYPE RECTIFIER

Rectifying diodes are the critical components in rectifiers, which are responsible for converting microwave power into direct current power due to their nonlinear properties. Typically, optimal efficiency is reached when the dc voltage reaches half the values of both the breakdown voltage and the forward voltage, as described below:

$$V_{DC} = (V_{br} + V_{th}) / 2 \quad (1)$$

where V_{DC} is the dc voltage, V_{br} is the breakdown voltage, and V_{th} is the forward voltage.

Rectifiers offering a broad dynamic range can maintain high efficiency at low and high power levels. Three approaches have been introduced to reduce the sensitivity of rectifying efficiency to fluctuations in input power: boosting the breakdown voltage of rectifying diodes [10][11][12][13], incorporating power recycling methods [14][15], and minimizing variations in input impedance resulting from changes in input power [16].

A high-efficiency rectifier featuring self-tuning impedance matching using a varactor was introduced by us, [16] as illustrated in Fig. 1(a). The varactor's capacitance is tuned based on the output dc voltage, which correlates with the input microwave power level. Additionally, the diode's impedance will fluctuate in response to changes in input power. Therefore, the varactor can be utilized to compensate for the variations in diode impedance resulting from these input power changes. Thus, a wide range of input microwave power is achieved.

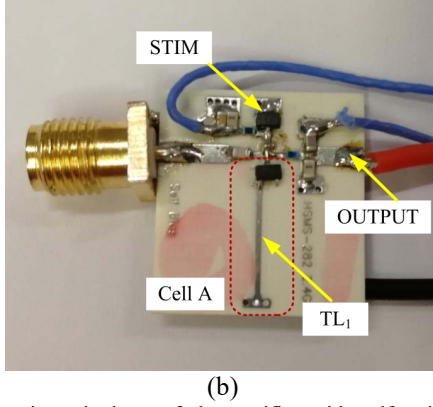
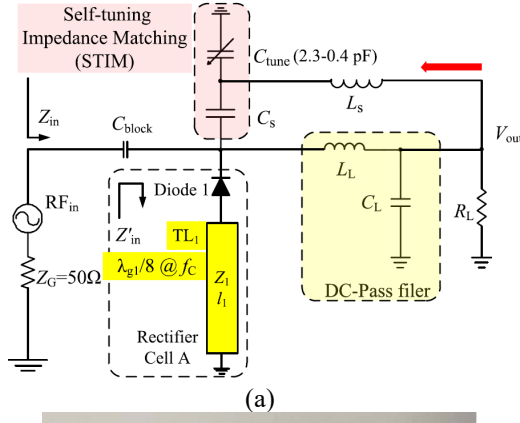


Fig. 1 Schematic and photo of the rectifier with self-tuning impedance matching

Fig. 1(b) shows the fabricated rectifier, and Fig. 2 shows the measured conversion efficiency. With the compensation from the varactor, the rectifier improves the microwave power dynamic range by 10%. The rectifier efficiency is higher than 50%, from 2.5 dBm to 25.5 dBm, with a dynamic range of 23 dB.

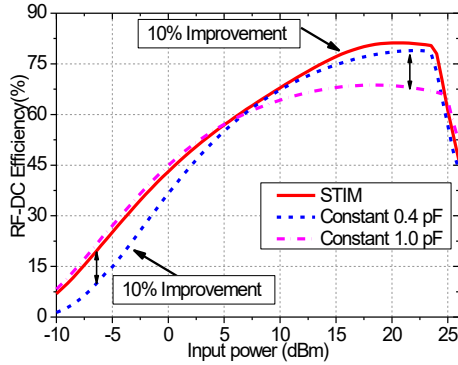


Fig. 2 Rectifying efficiency with respect to input microwave power.

III. ADAPTIVE TYPE RECTIFIER

We have two parallel nonlinear resistances, R_1 and R_2 , as shown in Fig. 2. The power distribution between them is dependent on their resistances:

$$\frac{P_1}{P_2} = \frac{R_2(P_2)}{R_1(P_1)} \quad (2)$$

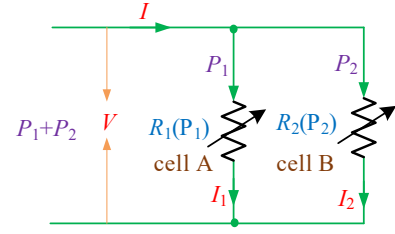
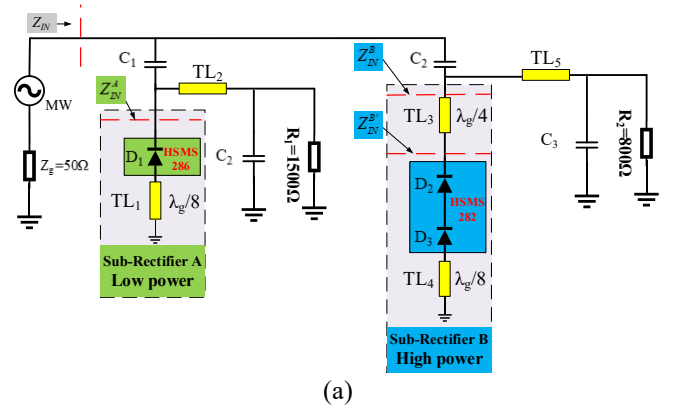


Fig. 3 Power distribution between two branches

With the nonlinear characteristics of the diode, input microwave power is distributed to the two branches adaptively, e.g., at low power case, R_1 is less than R_2 and achieves most of the power; at high power case, R_2 is less than R_1 and achieves most of the power. Thus, no matter at low or high power cases, rectifiers keep at the optimal status and achieve high efficiency in a wide input power range.

We introduced a wide dynamic range rectifier that utilizes two sub-rectifiers to accommodate both low and high power levels, as depicted in Fig. 4(a). The fabricated rectifier is shown in Fig. 4(b). By connecting two HSMS282 diodes in series, the breakdown voltage of the rectifying diode is doubled. This branch is applied to rectify high microwave power levels. The other branch uses an HSMS286 diode to rectify the low-power microwave. Thus, the rectifier may adaptively switch from one branch to the other based on the input microwave power level.

Measured and simulated results of the proposed rectifier are shown in Fig. 5. As a result, the power range in which the conversion efficiency exceeds 50% is expanded to the range from -1 dBm to 30 dBm (31 dB). [10] The rectifier uses three diodes to realize a wide dynamic input power range. The sub-rectifiers are designed for high and low power levels, respectively.



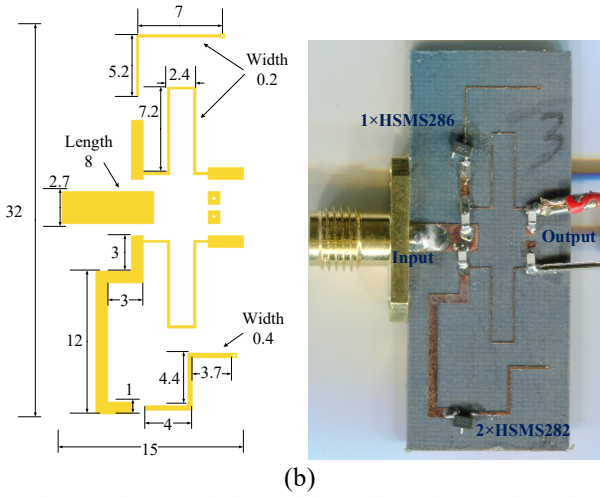


Fig. 4 Schematic, layout, and photo of the rectifier with two sub-rectifiers for various power levels.

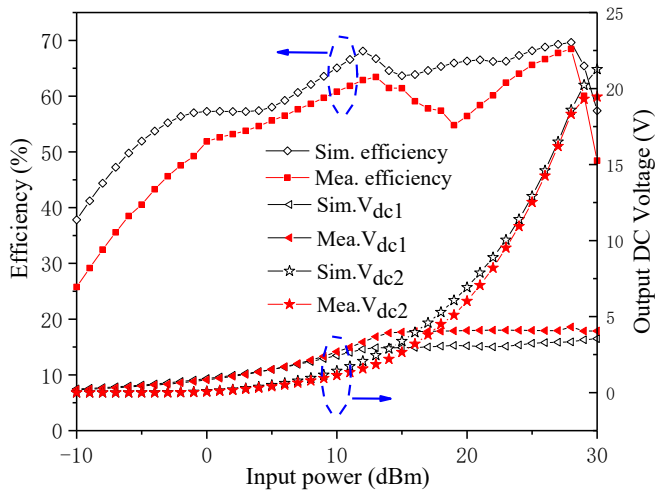


Fig. 5. Simulation and measurement results for the output dc voltage and RF-dc rectifying efficiency of the rectifier II at various input power levels.

IV. CONCLUSIONS

WPT develops fast and faces challenges from many practical fields, e.g., wireless power supply to movable objects. The received power levels greatly depend on the distance, beam angles, polarizations, etc. Thus, rectifiers with large dynamic power ranges are crucial to a WPT system and must be well-studied.

Two methods to design microwave rectifiers with large power dynamic ranges are presented. One is the feedback type rectifier, which feedbacks the output DC voltage to a varactor to control the impedance. It is a dynamic impedance compensation to achieve impedance matching with a large dynamic power range. The other is the adaptive strategy to distribute power automatically with the input power levels. The input power is adaptively distributed to two sub-rectifiers with respect to the input power level and keeps them in the high rectifying

efficiency status. Finally, a 31 dB of input microwave power dynamic range is achieved. In the future, high-efficiency rectifiers with large dynamic input power ranges will be applied to WPT systems.

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